Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Course : B.Tech
Code : 101407

Subject: Mechanical Engineering

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only): **[2 x 7 = 14]**

(a) The Zeroth Law of Thermodynamics is used to:
(i) Define entropy
(ii) Define pressure
(iii) Define temperature
(iv) Define enthalpy

(b) Which of the following is an extensive property of a system?
(i) Pressure
(ii) Temperature
(iii) Volume
(iv) Density

(c) A gas expands from 0.5 m^3 to 2 m^3 at constant pressure of 200 kPa. The work done by the gas is:
(i) 300 kJ
(ii) 400 kJ
(iii) 500 kJ
(iv) 600 kJ

(d) The Carnot cycle consists of:
(i) Two isothermal and two adiabatic processes
(ii) Two constant volume and two adiabatic processes
(iii) Two constant pressure and two adiabatic processes
(iv) Four adiabatic processes

(e) efficiency of a Carnot engine operating between 300 K and 600 K is:
(i) 33.3%
(ii) 50%
(iii) 66.6%
(iv) 75%

(f) A heat engine absorbs 1000 kJ from a reservoir at 800 K and rejects heat to a reservoir at 300 K. The maximum possible work output is:
(i) 400 kJ
(ii) 500 kJ
(iii) 625 kJ
(iv) 700 kJ

(g) The Rankine cycle is used in:
(i) Gas turbines
(ii) Steam power plants
(iii) Refrigerators
(iv) Internal combustion engines

(h) Which property is used to measure the randomness of a system?
(i) Enthalpy
(ii) Entropy
(iii) Work
(iv) Energy

(i) The entropy change (in kJ/kgK) of a system undergoing a reversible process where heat transfer is 500 kJ and temperature is 400 K is:
(i) 0.5
(ii) 1
(iii) 1.5
(iv) 1.25

(j) In an ideal gas mixture, the total pressure is given by:
(i) Boyle's law
(ii) Avogadro's law
(iii) Dalton's law
(iv) Charles' law

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Q.2 (a) A system contains 2 kg of an ideal gas at a pressure of 200 kPa and a temperature of 300 K. The gas undergoes a quasi-static expansion following the relation PV = constant until the volume is doubled. Calculate the final temperature and work done during the process. [7]

(b) Explain the concept of entropy and its significance in thermodynamic processes. A heat engine operates between 600 K and 300 K reservoirs. Determine the maximum possible efficiency of the engine. [7]

Q.3 (a) Describe the Carnot cycle with a neat diagram and derive the efficiency expression for an ideal Carnot heat engine. A Carnot engine receives 500 kJ of heat from a high-temperature reservoir at 1000 K and rejects heat to a low-temperature reservoir at 300 K. Determine the work output and heat rejected. [7]

(b) Explain the Rankine cycle with the help of a T-S diagram. Steam enters the turbine at 6 MPa and 600°C and exhausts at 10 kPa. Determine the thermal efficiency of the cycle assuming ideal conditions. [7]

Q.4 (a) Explain the concept of exergy and irreversibility in thermodynamic processes. A heat exchanger receives 1500 kJ of heat at 500 K and rejects it at 300 K. Determine the exergy destruction during the process. [7]

(b) Explain the concept of exergy and irreversibility in thermodynamic processes. A heat exchanger receives 1500 kJ of heat at 500 K and rejects it at 300 K. Determine the exergy destruction during the process. [7]

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Q.5 (a) Derive the steady-flow energy equation and apply it to a nozzle. Explain how the velocity of the fluid changes in a convergent nozzle. [7]

(b) State and explain Kelvin-Planck and Clausius statements of the second law of thermodynamics. Prove that violation of one statement leads to the violation of the other. [7]

Q.6 (a) Derive the Maxwell's relations from the fundamental thermodynamic equations. A gas expands from an initial state of 500 kPa and 0.1 m^3 to a final state of 100 kPa in a polytropic process with $n = 1.3$. Calculate the work done and the heat transfer in the process. [7]

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(b) A gas undergoes a polytropic process (PV^n = constant) from an initial state of 500 kPa and 0.1 m^3 to a final volume of 0.5 m^3. If n = 1.3, determine the final pressure. [7]

Q.7 (a) Explain the Brayton cycle with a T-S diagram and derive an expression for its efficiency. [7]

(b) A gas turbine operates on an ideal Brayton cycle between pressure limits of 100 kPa and 800 kPa. The temperature at the compressor inlet is 300 K and the maximum cycle temperature is 1400 K. Determine the thermal efficiency and work ratio. [7]

Q.8 (a) A psychrometric air-conditioning system processes 100 m^3/min of air at 35°C and 60% relative humidity. The air is first cooled and dehumidified to 18°C using a cooling coil and then reheated to 22°C before being supplied to a room. Determine the cooling load and reheating load required [7]

(b) Working principle of an absorption refrigeration system and compare it with a vapor compression refrigeration system. [7]

Q.9 Write short notes on the following:-

(a) Concept of thermodynamic equilibrium and its significance in engineering applications. [7]

(b) Clausius-Clapeyron equation and its applications in phase change calculations. [7]

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